

An Evidence-Driven ePolyScat Research Portal

End-to-End Scientific Workflows, Reproducible Inputs, and Verified Computation

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PROJECT ACCEPTANCE
REVIEW
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Scientific Gateway Delivery

RESEARCH GAP

Scientific workflows span specialized programs, remote schedulers, and local analysis tools. Job completion alone does not prove that scientifically valid outputs were produced or that the correct files reached the next stage.

CONTRIBUTION A · ZIMING ZHANG Workflow Portal and Scientific Verification

KEY CONTRIBUTION

We transformed ePolyScat from a collection of disconnected tools into a guided, inspectable workflow. Each stage preserves the researcher's original input, records output provenance, and exposes a verified handoff to the next calculation or visualization step.

Evidence narrative: from input to interpretation

01 Configure

it Run on 2026...

Modules

Utilities

Workflows

1 Data Generation

2 ePolyScat Run

3 Post-processing

4 Visualization

Input Computation

Gaussian16

OpenMolcas

Required Files

Molcas_input

Select a complete four-stage workflow and make required scientific inputs explicit.

02 Preserve

it Run on 2026...

Data Entry

Grid / Expansion

State Definitions

Potentials

Energies / Partial Waves

Outputs

Ordered Sequence

1 LMax

Edit structured parameters without losing commands, comments, repeated records, or source ordering.

03 Verify

EPolyScat Run on 2026/7/19 09:14:18

Modules/EPOLYSCAT_DMAT

Continue in Workflow

Visualzation & Analysis

Manual selection required

Input

epolyscat_ch4_test

epolyscat_ch4_test

Output

epolyScat.stderr

epolyScat.stdout

Separate scheduler completion from scientific success using application-specific output evidence.

04 Interpret

Continue verified results into post-processing and local scientific visualization.

Engineering decisions

- Lossless ordered document model for reproducible ePolyScat inputs.
- Semantic output roles instead of filename-only inference.
- Resource-aware execution contracts across supported HPC systems.
- Provenance-aware continuation with researcher review before submission.

Scientific value

Researchers can reproduce exactly what was submitted, distinguish scheduler artifacts from scientific results, and continue a calculation without manually rebuilding file dependencies.

Reproducible input → interpretable output

ACCEPTANCE RESULT

Portal configuration → HPC execution → scientific proof → provenance-aware handoff → local visualization is operational and demonstrated with real runs.

Acceptance evidence

296

Automated Checks Passed

Non-gRPC test coverage, HPC wrapper regressions, and the production build passed successfully.

3

HPC Platforms Validated

2

Scientifically Verified Runs

4

Operational Workflow Stages

ACCEPTANCE CLAIM

Connect configuration, execution, scientific validation, and stage-to-stage handoff in one portal experience.

CONTRIBUTION B · NGUYEN, THINH M [Partner Progress Title]

[One sentence: what changed, for whom, and why it matters.]

Problem & method

[Problem and method: summarize the approach in two or three concise points.]

Implementation

[Implementation or experimental contribution: name the system, artifact, or analysis delivered.]

Primary result

PRIMARY RESULT FIGURE

Replace \PartnerVisual with
\includegraphics[width=\linewidth]{...}

[One sentence explaining what the visual proves.]

Validation evidence

✓ [Tests, measurements, comparison, or user evidence supporting the claim.]

Outcome metrics

[Two to four defensible outcome metrics, separated by centered dots.]

How this contribution fits the shared objective

01

Traceable configuration

02

Verified computation

03

Reviewable research output

REPLACEMENT GUIDE Edit the eight \Partner... macros near the top; every labeled area updates automatically.